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Self-medication practices for COVID-19: a systematic review

--Manuscript Draft--

Manuscript Number:	PONE-D-21-21439
Article Type:	Research Article
Full Title:	Self-medication practices for COVID-19: a systematic review
Short Title:	Self-medication practices for COVID-19
Corresponding Author:	Alvaro Taype-Rondan Universidad San Ignacio de Loyola Lima, PERU
Keywords:	COVID-19; SARS-CoV-2; Self Medication; Global Health; Pharmaceutical Preparations; Systematic Review
Abstract:	Objectives As the COVID-19 pandemic progresses, many people may embrace self-medication, whose frequency and characteristics are important subjects of study in order to understand its dangers and design interventions. We conducted a systematic review to assess the self-medication frequency for the prophylaxis or management of COVID-19. Methods For this systematic review, we used different keywords and searched studies published in PubMed, Scopus, Web of Science, Embase, two preprints repositories, Google, and Google Scholar. We included studies which reported original data and assessed self-medication for prophylaxis of COVID-19 or management as main or secondary outcome. The risk of bias assessment was evaluated with the Newcastle-Ottawa Scale modified for cross-sectional studies. Results We found 11 studies that assessed self-medication for prevention or management of the COVID-19. Most of the studies did not detailed the questions used to assess self-medication nor the recall period. In the 7 studies that assessed self-medication for any reason, the self-medication frequencies ranged between 14.8% and 88.3%, and the most mentioned drugs were antibiotics, acetaminophen, chloroquine or hydroxychloroquine, vitamins, or supplements, ivermectin, ibuprofen, and antiasthmatics. Most of the studies had a risk of bias in the representativeness of the sample and the outcome assessment. Conclusions Self-medication frequency was heterogeneous, and the methods used in the included studies were unclear. Thus, more well-designed, and adequately reported studies in this regard are warranted.
Order of Authors:	Alvaro Quincho-Lopez
	Christeam A Benites-Ibarra
	Maryori M. Hilario-Gomez
	Renatta Quijano-Escate
	Alvaro Taype-Rondan
Additional Information:	
Question	**Response**
Financial Disclosure	The author(s) received no specific funding for this work.

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Short title: Self-medication practices for COVID-19

Authors:

Alvaro Quincho-Lopez[1]

Christeam A. Benites-Ibarra[2]

Maryori M. Hilario-Gomez[1]

Renatta Quijano-Escate[3]

Alvaro Taype-Rondan[4]

Affiliations:

[1]Universidad Nacional Mayor de San Marcos, Sociedad Científica de San Fernando, Lima, Peru

[2]Sociedad Científica de Estudiantes de Medicina de la Universidad Nacional del Santa, Nuevo Chimbote, Ancash, Peru

[3]Universidad Nacional San Luis Gonzaga, Sociedad Científica de Estudiantes de Medicina de Ica, Ica, Peru

[4]Universidad San Ignacio de Loyola, Unidad de Investigación para la Generación y Síntesis de Evidencias en Salud, Lima, Peru

Funding:

This study was self-funded by the authors.

Corresponding author:

Alvaro Taype-Rondan

Universidad San Ignacio de Loyola, Unidad de Investigación para la Generación y Síntesis de Evidencias en Salud

Av. La Fontana 750, Lima, Peru

alvaro.taype.r@gmail.com

Abstract

Objectives: With the progression of the coronavirus disease 2019 (COVID-19) pandemic, several individuals may adopt self-medication. To understand the associated dangers and design interventions, it is important to study the ~~frequency~~ prevalence and characteristics of self-medication. We conducted a systematic review to assess the ~~frequency~~ of self-medication for the prophylaxis or management of COVID-19.

Methods: For this systematic review, we used different keywords and searched studies published in PubMed, Scopus, Web of Science, Embase, two preprint repositories, Google, and Google Scholar. We included studies that reported original data and assessed self-medication for the prophylaxis or management of COVID-19 as a main or secondary outcome. Risk of bias was assessed using the Newcastle–Ottawa Scale modified for cross-sectional studies.

Results: We identified 11 studies that assessed self-medication for the prevention or management of COVID-19. Most studies ~~neither~~ detailed the questions used to assess self-medication nor the recall period. In the seven studies that assessed self-medication for any reason, frequency of self-medication ranged between 14.8% and 88.3%, and the most mentioned drugs were antibiotics; acetaminophen; chloroquine or hydroxychloroquine; vitamins or supplements; ivermectin, ibuprofen, and antiasthmatics. Most studies had a risk of bias in the representativeness of the sample and assessment of outcome.

Conclusions: The frequency of self-medication was heterogeneous, and the methods used in the included studies were unclear. Thus, more well-designed and adequately reported studies in this regard are warranted.

Keywords: COVID-19, SARS-CoV-2, Self-Medication, Global Health, Pharmaceutical Preparations, Systematic Review

1 Introduction

According to the World Health Organization (WHO), self-medication is the administration of medications without consulting a health professional and without medical supervision [1]. Its prevalence varies with country [2–4], age group [5,6], and occupation [7].

With the progression of the coronavirus disease 2019 (COVID-19) pandemic, several drugs have been proposed as potential candidates for this disease [8], most of which are now known to have little or no benefit [9,10]. However, the fear of contracting the virus, low access to health services, and massive misinformation has led some people to employ self-medication, with expected direct and indirect consequences [11,12].

Previous studies have assessed the frequency and characteristics of self-medication for COVID-19 and have helped understand which useless or potentially hazardous drugs are being employed and which characteristics predispose individuals to a greater frequency of self-medication. Thus, in this systematic review, we aimed to assess the frequency of self-medication for the prophylaxis or management of COVID-19.

2 ~~Materials and~~ methods

This systematic review was reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (Supplement Table S3) [13]. The study protocol has been registered at PROSPERO (CRD42021236191).

2.1 Data sources

On February 4th, 2021, we searched PubMed, Scopus, Web of Science, Embase, two preprint repositories (MedRxiv and SciELO Preprints), Google, and Google Scholar. No language restriction was applied. The search was restricted to include documents published in 2020 or 2021 (see S1 Table).

2.2 Inclusion criteria

We included original studies, either published in scientific journals or with full text available in preprint repositories, which reported original data and assessed the use of self-medication for the prophylaxis and/or management of COVID-19 as a main or secondary outcome. Self-medication was defined as the use of ~~drugs~~ without a prescription from a health professional.

2.3 Study selection

Two authors (AQL and MHG) independently selected the studies. Screening included a review of the search results by title and abstract according to the inclusion criteria using the web program Rayyan (http://rayyan.qcri.org), followed by reviewing the full text of the relevant studies to determine whether they were appropriate for study inclusion. Discrepancies were consulted with another author (ATR) and resolved with consensus.

2.4 Data extraction

Two authors (CBI and RQE) independently extracted the relevant data for the review using a prepiloted Microsoft Excel spreadsheet. Again, any discrepancies were discussed with another author (ATR) and resolved with consensus. The following variables were assessed for each study: author, year of publication, country, study design, setting, population, sample technique, date of data collection, number of participants, age, sex, characteristics that may be related to self-medication, method of assessment of self-medication, results of frequency of self-medication, reasons for practicing self-medication (including who recommended such medication), from where the medication was obtained, and adverse events related to its use.

2.5 Risk-of-bias assessment

Two authors (CBI and RQE) independently evaluated the risk of bias for each included study using the Newcastle–Ottawa Scale (NOS) adapted for descriptive cross-sectional studies [14]. This scale comprises two criteria: the selection criteria (representativeness of the sample, sample size, and nonresponders) and the outcome criteria (outcome assessment and statistical tests). Any discrepancy was consulted with another author (ATR).

2.6 Statistical analysis

Because studies were performed in different populations and using different definitions of self-medication, we did not perform meta-analyses and decided to present the results of each study separately.

3 Results

3.1 Study selection

A total of 765 records were found in the database search. After the duplicates were eliminated, we screened 227 records, of which we reviewed 46 full-text documents. We excluded 35 articles (S2 Table) and included 11 articles, all of which were cross-sectional studies (Fig. 1).

[Insert Figure 1]

Fig 1. Flow diagram of study selection.

3.2 Methodological quality of the included studies

In the risk-of-bias assessment using the NOS, most studies received 4 out of 5 stars; all studies except one [15] failed in the representativeness of the sample item of the selection criteria (Table 1). We considered the article by Sadio et al twice because of the different sampling methods used in that study (probability and nonprobability).

Table 1. Quality assessment of included studies using the Newcastle–Ottawa Scale (NOS) adapted for cross-sectional studies.

Study	Final score[6]	Representativeness of the sample[1]	Sample size[2]	Not respondents[3]	Outcome evaluation[4]	Statistical test[5]
Quispe-Cañari (2020)	4/5		★	★	★	★
Nasir (2020)	1/5				★	
Wegbom (2021)	3/5		★	★		★
Dare (2021)	2/5			★		★
Ahmed (2020)	0/5					
Mansuri (2020)	4/5		★	★	★	★

Zavala-Flores (2020)	1/5				★	
Sadio (2021)[7]	5/5	★	★	★	★	★
Sadio (2021)[8]	4/5		★	★	★	★
Miñan-Tapia (2020)	4/5		★	★	★	★
Kamarli Altun (2020)	2/5		★	★		★
Chauhan (2020)	0/5					

[1]Representativeness of the sample: One star was awarded to studies with random sampling.
[2]Sample size: One star was awarded to studies with a justified and satisfactory sample size.
[3]Non-respondents: If comparability between respondents' and nonrespondents' characteristics was established and the response rate was satisfactory, one star was assigned.
[4]Outcome evaluation: If the study explicitly defined how self-medication was defined and how long the recall period was, one star was assigned.
[5]Statistical test: One star was assigned if it was not a complex sample and the sample had been calculated correctly or if it was a complex sampling and the sampling was considered to determine the prevalence.
[6]The sum of stars of 1+2+3+4+5
[7]Probability sampling.
[8]Nonprobability sampling.

3.3 Study characteristics

The 11 included studies reported data on self-medication practices in eight different countries: three from Peru [16–18], two from Bangladesh [19,20], one from India [21], one from Togo [15], one from Nigeria [22], one from Uganda [23], one from Turkey [24], and one from the Kingdom of Saudi Arabia [25]. All studies were cross-sectional. Most of the studies (N = 7) were published as original reports, two were letters to the editor [17,21], and two were preprints reporting original data [18,23]. Studies were performed from March to August 2020.

Six studies were conducted targeting the general population, whereas one study was conducted in hospitalized patients with COVID-19 [17], one in health science students [18], one in health workers [21], one in dietitians [24], and one in workers from five sectors (health care, air transport, police, road transport, and informal) [15].

The 11 studies comprised 125–3,792 participants. All studies were performed in adults. Eight studies mentioned the mean or median age of the patients, which ranged from ~~21 to 60.3~~ years. The percentage of females ranged from 28.4% to 88.2% (Tables 2 and 3).

Table 2. Characteristics and findings of studies that assessed self-medication of any drug for any reason.

Study	Country (city)/ date	Subjects (age, sex)	Survey application and self-medication definition	Frequency of self-medication for COVID-19 prevention or management (overall and most-used drugs)
Studies performed in the general population:				
Quispe-Cañari (2021)	Perú/June 2020	Adults (median age: 23 years, female: 54.5%), N = 3792	• Open online survey (convenience sampling) • Respondents were asked to indicate whether they consumed any of the following drugs: acetaminophen, ibuprofen, azithromycin, hydroxychloroquine, penicillin, or antiretrovirals (for any reason) • Recall period: during the COVID-19 lockdown	• **Overall:** 33.4% for any reason • **For any reason:** acetaminophen (27.0%), ibuprofen (7.4%), azithromycin (4.8%), penicillin (2.3%), antivirals (1.6%), hydroxychloroquine (0.7%)
Nasir (2020)	Bangladesh (Dhaka city)/from April to June 2020	Adults (age: 50.5% between 45 and 54 years, female: 55.3%), N = 626	• Open online survey (convenience sampling) • Self-medication was defined as taking drugs without prescription • Recall period: during the COVID-19 outbreak	• **Overall:** 88.3% for any reason • **For COVID-19 management:** ivermectin (77.2%), azithromycin (54.2%), montelukast (43.1%), calcium supplements (41.4%), doxycycline (40.3%), hydroxychloroquine (20.4%)
Wegbom (2021)	Nigeria/from June to July 2020	Adults (mean age: 42.2 years, female: 57.1%), N = 461	• Open online survey (convenience sampling) • Self-medication was defined as taking drugs intended to prevent/treat COVID-19 without prescription by medically qualified personnel (for either prevention or management) • Recall period: not detailed	• **Overall:** 41% for prevention or management • **For COVID-19 prevention or management:** Vitamin C or multivitamin (21.3%), antimalarial drugs other than hydroxychloroquine and chloroquine (19.3%), amoxicillin (10.2%), ciprofloxacin (6.1%), herbal products (4.1%), metronidazole (3.5%), erythromycin (2.2%), hydroxychloroquine or chloroquine (1.3%)
Ahmed (2020)	Bangladesh/from June to July 2020	Adults (mean age: 30.8 years, female: 38.6%), N = 1222	• In-person and online survey (purposive sampling to include participants of all ages and diverse educational and socioeconomic backgrounds) • Self-medication defined as the use of medicines and herbal products/foods as preventive and curative measures against COVID-19 • Recall period: not detailed	• **Overall:** 14.8% for prevention or management • **As a preventive measure:** arsenicum album (4.5%), vitamin supplements (4.0%), mineral supplements (2.9%), acetaminophen (2.4%), antihistamines (1.7%), antiasthmatics (1.3%), ivermectin (0.8%) • **To manage symptoms:** acetaminophen: (13.6%), antihistamines (6.0%), antibiotics (3.9%), mineral supplements (3.9%), antiasthmatics (3.6%), vitamin supplements (3.0%), ivermectin (2.2%)
Dare (2020) - preprint	Uganda (Western region)/from June to August 2020	Adults from Uganda western cities (age: 43.8% between 25 and 34 years, female: 45.2%), N = 272	• In-person and online survey (convenience sampling) • Self-medication assessment was not detailed • Recall period: during COVID-19 pandemic	• **Overall:** 57% for any reason
Studies performed in specific populations				
Sadio (2021)	Togo (Lomé)/from April to May 2020	Adults from any of these five sectors: health care, air transport, police, road transport, and informal (median age: 36 years, female: 28.4%), N = 955	• In-person survey through recruitment, open invitation, and random sampling (two or three stages) • Self-medication was defined as the use of medicines/drugs alleged to treat or prevent COVID-19 without a physician order • Recall period: 2 weeks	• **Overall:** 34.2% for prevention or management • Vitamin C (27.6%), traditional medicine (10.2%), chloroquine/hydroxychloroquine (2.0%), azithromycin (1.2%)
Miñan-Tapia (2020) - preprint	Peru (Tacna)/from June to July 2020	Undergraduate students of health-related careers from two universities (median age: 21 years, female: 69.1%), N = 718	• Online, convenience sampling • Self-medication was defined as using any of the prespecified 14 medicines for COVID-19 without medical prescription • Recall period: during the past 3 months	• **Overall:** 51.3% for any reason • Acetaminophen (41.4%), ibuprofen (21.1%), dexamethasone (10.7%), aspirin (8.6%), azithromycin (4.9%)

3.4 Definition of self-medication

All studies assessed self-medication through self-report. However, the questions used to assess self-medication were not explicitly mentioned in 9 of the 11 studies. Moreover, four studies did not clearly detail the recall period for self-medication: it was considered as anytime during the COVID-19 pandemic in four studies, in the past 2 weeks in one study, in the past 3 months in one study, and since the COVID-19 diagnosis in another study that assessed hospitalized patients (Tables 2 and 3).

3.5 Frequency of self-medication and most frequently used drugs

Seven studies assessed self-medication for any reason: five in the general population (with a range of frequency of self-medication between 14.8% and 88.3%) and two in specific populations (with a frequency of self-medication of 34.2% in adults who worked in any of the five assessed sectors and 51.3% in undergraduate students of health-related careers). Of these seven studies, only two specified which question was asked [16,18], whereas the other five studies did not detail whether participants could report self-medication only for a prespecified list of drugs or whether it was an open question.

Of the seven studies that assessed self-medication for any reason, six detailed the list of drugs reported by the participants. The reported drugs were different across the six studies: six studies reported the consumption of antibiotics; three of acetaminophen; four of chloroquine or hydroxychloroquine; four of vitamins or supplements; two of ivermectin; two of ibuprofen; two of antiasthmatics; and one of other drugs. In addition, consumption preferences varied across studies. For example, when drugs were sorted in the order of highest to lowest consumption, vitamins/supplements occupied the first place in two studies, fourth place in one study, and in the last study, second place as a preventive measure and sixth place to manage symptoms (Table 2).

The other four studies assessed self-medication of specific drugs or for specific reasons: one of these studies was performed in the general population of Saudi Arabia and found that 35.1% of the surveyed individuals used self-medication for fever [25]. Another study found that 33.9% of patients with COVID-19 in a Peruvian hospital used self-medication since their diagnosis (mainly with antibiotics, ivermectin, and corticosteroids) [17]. The other two studies assessed the use of a specific

drug for COVID-19 prevention: one study found that 94.5% of the surveyed dietitians in Turkey consumed dietary supplements for COVID-19 prevention [24], whereas the other study found that 46.4% of the surveyed doctors and nurses in India used hydroxychloroquine for the prevention of COVID-19 [21] (Table 3).

Table 3. Characteristics and findings of studies that assessed self-medication of specific drugs or for specific reasons.

Study	Country (city)/ date	Subjects (age, sex)	Survey application and self-medication definition	Frequency of self-medication for COVID-19 prevention or management (overall and most-used drugs)
Studies that assessed the use of drugs for specific reasons:				
Mansuri (2020)	Kingdom of Saudi Arabia/from March to April 2020	Adults (age: 60.8% <40 years, female: 60.3%), N = 388	• Online electronic survey, convenience sampling • Self-medication was assessed using the question, "Are you self-medicating for fever?" • Recall period: during COVID-19 pandemic	• **Overall:** 35.1% to manage fever
Zavala-Flores (2020)– letter to the editor	Peru (Lima)/June 2020	Hospitalized adults with COVID-19 (Mean age: 60.3, Female: 30.3%), N = 132	• Authors included all patients hospitalized during a certain week • Self-medication was defined as using some type of COVID-19 medication before hospitalization without medical prescription • Recall period: since the COVID-19 diagnosis	• **Overall:** 33.9% for management • Antibiotics (28.3%), ivermectin (20.7%), corticosteroids (17.0%), acetaminophen (12.3%), aspirin (4.7%), NSAID (4.7%), hydroxychloroquine (0.9%), others (1.9%)
Studies that assessed the use of specific drugs:				
Kamarli Altun (2020)	Turkey (Antalya)/from May to June 2020	Dietitians graduated from Turkish universities (mean age: 30.6 years, female: 88.2%), N = 550	• Online, sample with a known universe • Self-medication was defined as taking dietary supplements for COVID-19 prevention • Recall period: not detailed	**Overall:** 94.5% took dietary supplements for COVID-19 prevention
Chauhan (2020)– letter to the editor	India/dates not specified	Doctors and nurses (mean age: 37.7 years, female: 29.6%), N = 125	• Online, voluntarily completed • Self-medication was defined as taking hydroxychloroquine for COVID-19 prevention • Recall period: not detailed	**Overall:** 46.4% used hydroxychloroquine for COVID-19 prevention

NSAID, nonsteroidal anti-inflammatory drug

3.6 Reasons to practice self-medication

Five of the 11 studies mentioned the reasons for self-medication. However, this was assessed heterogeneously across studies. Thus, one study referred to the symptoms, another suggested that participants used self-medication based on the recommendations of national groups, and the other three studies mentioned a plethora of reasons, including the fear of stigmatization, fear of quarantine,

affordability, convenience of self-medication, patients believing that the symptoms were not serious, and suggestions of family and friends (Table 4).

3.7 Who suggested the practice of self-medication and from where were the medications obtained?

Three studies assessed who suggested the use of self-medications. One study reported that patients received advice from family, friends, or relatives (40.1%) or relied on their own knowledge (32.4%). Media (print, digital, and social) was also a major source of information. However, only 25% of the participants sought advice from doctors, whereas 18% consulted with pharmacists [19]. Another study reported that patients were advised by a doctor, pharmacist, literature, medicine company, friends, or TV/radio programs [24]. The third study indicated that patients relied on the Indian Council of Medical Research guideline (71.4%) or advice from a senior fellow or colleague (23.8%) [21].

Only one study [22] described from where the patients obtained their self-medications. The most common sources for obtaining medications were the pharmacy (73.9%), patent medicine vendor (23.6%), hospital (7.6%), hawkers (4.5%), and faith-based outlets and herbalists (2.1%, each).

Table 4. Reasons to practice self-medication, who suggested its practice, and from where such medications were obtained.

Study	Reasons to practice self-medication	Suggested by	Obtained from
Nasir (2020)	Fever (37.6%), throat pain (28.8%), dry cough (14.2%), loss of smell (9.2%), loss of taste (3.5%), body ache (5.0%), and rarely diarrhea (1.7%)	NR	NR
Chauhan (2020)	The article suggested that self-medication was based on the recommendations made by the Indian Council of Medical Research on the empirical use of hydroxychloroquine	Indian Council of Medical Research guideline (71.4%), advice from senior fellow or colleague (23.8%)	NR
Wegbom (2021)	Fear of stigmatization or discrimination (79.5%), fear of quarantine or self-isolation (77.3%), fear of infection or contact with a suspected or known COVID-19 patient (76.3%), delay in receiving treatment at health facilities" (55.6%), influence of friends" (55.2%), social media (54.3%), nonavailability of drugs for COVID-19 treatment in the health facilities (53%), emergency illness (49.1%), delay in getting hospital services (28.1%), distance to the health facility (23%), proximity of the pharmacy (21%), nonavailability of medicine in a health facility (19.3%), and health facility charges (15.3%)	NR	Pharmacy (73.9%), patent medicine vendor (23.6%), hospital (7.6%), hawkers (4.5%), faith-based outlets and herbalists (2.1%, each)

Dare (2020)	Self-medication is affordable (37%) and convenient (32%), lack the means to get to the health facility/hospital (15%), fear of being diagnosed COVID-19 positive (9%), fear of visiting health facility or hospital (7%)	NR	NR
Miñan-Tapia (2020)	Believing that the symptoms were not so serious to go to a doctor (64.3%), referring to family/friends that are nonmedical health professionals (34.9%), and owing to economic reasons or use of over-the-counter medications (34.9%)	NR	NR
Ahmed (2020)	NR	Advice of family, friends, or relatives (40.1%) and their own knowledge (32.4%); advice from doctors (25%); advice from pharmacists (18%)	NR
Kamarli Altun (2020)	NR	Advise of a doctor, pharmacist, literature, medicine company, friends, or TV/radio programs	NR

NR, not reported

3.8 Adverse events due to self-medication for COVID-19

Only three studies reported whether adverse events occurred. One study found that 11.7% patients presented with side effects related to self-medication in the past 3 months [18]. Nonetheless, this study did not specify the adverse effects that occurred. The second study reported body rash (23.1%) followed by worsened condition (17.3%), yellowish eyes (7.7%), swollen face (3.8%), and vomiting of blood and severe diarrhea (5.8%) [22]. The third study reported only gastrointestinal upset [21].

4 Discussion

Because self-medication is a global phenomenon that may involve health risks at both the individual and community level, it is important to assess its frequency [26,27]. Previous studies have found that self-medication is a common practice, with a prevalence of 53%, 44%, and 53.6% (recall period: a single day to 6 months) according to systematic reviews conducted in Iran [2], Ethiopia [3], and India [4], respectively.

Although studies were performed in eight different countries, the lack of studies in some regions such as North America, the Middle East, North Africa, and Oceania is evident. It is expected that the intercountry differences in drug promotion, regulations, and the possibility of accessing some drugs

without a prescription influence the self-medication patterns [28,29]. In this review, 10 of the 11 studies were performed in low- and middle-income countries. However, the results do not suggest that self-medication is higher in low- and middle-income countries than in other countries.

Some of the included studies assessed symptomatic drugs such as acetaminophen or nonsteroidal anti-inflammatory drugs, which are some of the symptomatic medications used for COVID-19, along with several other diseases. In addition, studies have found an important, although heterogeneous, frequency of self-medication use with antibiotics, chloroquine or hydroxychloroquine, vitamins or supplements, ivermectin, and antiasthmatics. As stated by the current guidelines, none of these drugs has proven to have any benefit in the prevention or the ambulatory management of COVID-19 [30,31].

In fact, the adverse effects of some of these drugs are of great concern, such as antibiotic resistance due to extensive antibiotic use, bleeding caused by aspirin use, inhibition of the immune system caused by corticosteroids, or arrhythmia caused by hydroxychloroquine. Some healthcare organizations have issued statements on self-medication. For instance, the WHO recognizes that "successful" (i.e., beneficial) self-medication can be achieved in many countries only by improving individuals' knowledge and education level in such a way that the potential damages of this practice can be avoided [1]. Similarly, the International Pharmaceutical Federation, in tandem with the World Self-Medication Industry [32] as well as the World Medical Association [33], emphasizes the responsible use of nonprescription drugs.

As found in previous systematic reviews [5,6], studies tended to use different questions to assess self-medication or they simply did not specify the question used. Moreover, studies have established different recall periods, and even those that assessed self-medication since the beginning of the pandemic would have different time frames depending on the month in which they were performed. This prevents direct comparisons between studies and the subsequent meta-analysis of their results [34], which adds a potential source of bias that affects the internal validity of the results [34,35].

Most studies seemed to have been performed using convenience sampling. This may lead to misleading results because the results cannot be extrapolated to the general population owing to the nonprobability sampling used as well as the risk of sampling and nonresponder biases [36]. Accordingly, in this review, the representativeness of the sample domain was the worst-rated item in the risk-of-bias assessment.

5 Conclusion

We identified 11 studies that assessed the use of self-medication for the prevention or management of COVID-19. Most studies neither detailed the questions used to assess self-medication nor the recall period. In the seven studies that assessed self-medication for any reason, the frequencies of self-medication ranged from 14.8% to 88.3%, and the most mentioned drugs were antibiotics; acetaminophen; chloroquine or hydroxychloroquine; vitamins or supplements; ivermectin; ibuprofen; and antiasthmatics. Most studies had a risk of bias in the representativeness of the sample and assessment of the outcomes. Thus, more well-designed and adequately reported studies are warranted in this regard.

Acknowledgments

None.

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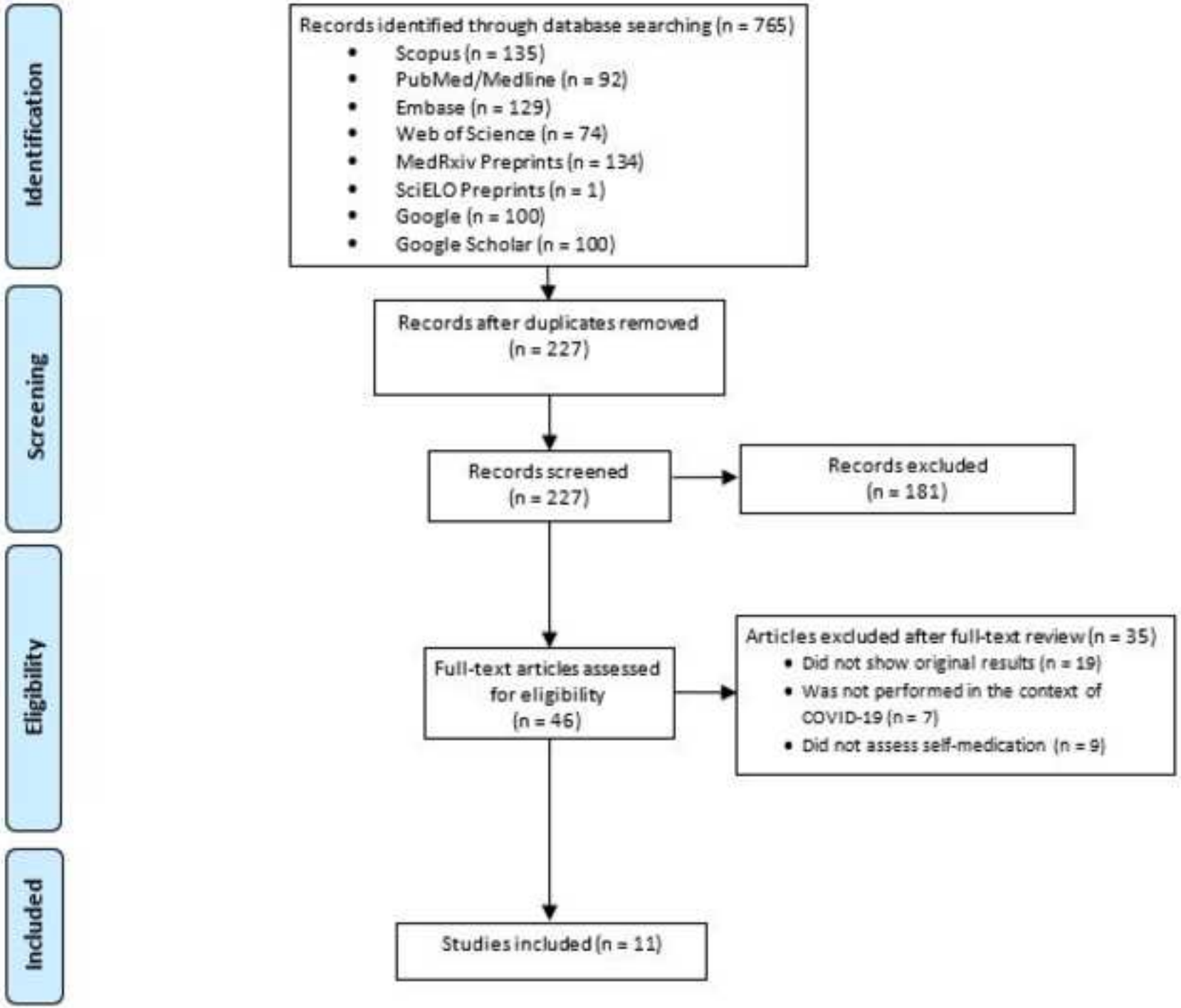
S1 Checklist. PRISMA 2009 checklist.

S1 Table. Search strategies.

S2 Table. List of articles excluded after full-text review.

Figure 1

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S1 Table

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